

# Static and Dynamic System

| S No. | Parameter        | Static System                        | Dynamic System                         |
|-------|------------------|--------------------------------------|----------------------------------------|
| 1.    | Explanation      | System does not change over time.    | System changes over time.              |
| 2.    | I/O Relationship | Fixed Input Output Relationship      | Time Varying Input Output Relationship |
| 3.    | Representation   | Represented by algebraic equations   | Represented by differential equations. |
| 4.    | Time Invariance  | Behaviour does not change over time. | Behaviour changes over time.           |
| 5.    | Behaviour        | Linear and predictable pattern       | Non Linear and non predictable pattern |



**Julen Urain** 

4 years ago

Very elegant approach. It would be nice to point out why the Hamiltonian approach is better for modelling in comparison with the baseline

Ref: [Static and dynamic system difference and information](#)

# Coupled Dynamical System

(LV) Dynamical system  
of Prey-Predator Model

- $\frac{dR}{dt} = rR - \alpha FR$
- $\frac{dF}{dt} = \beta FR - bF$

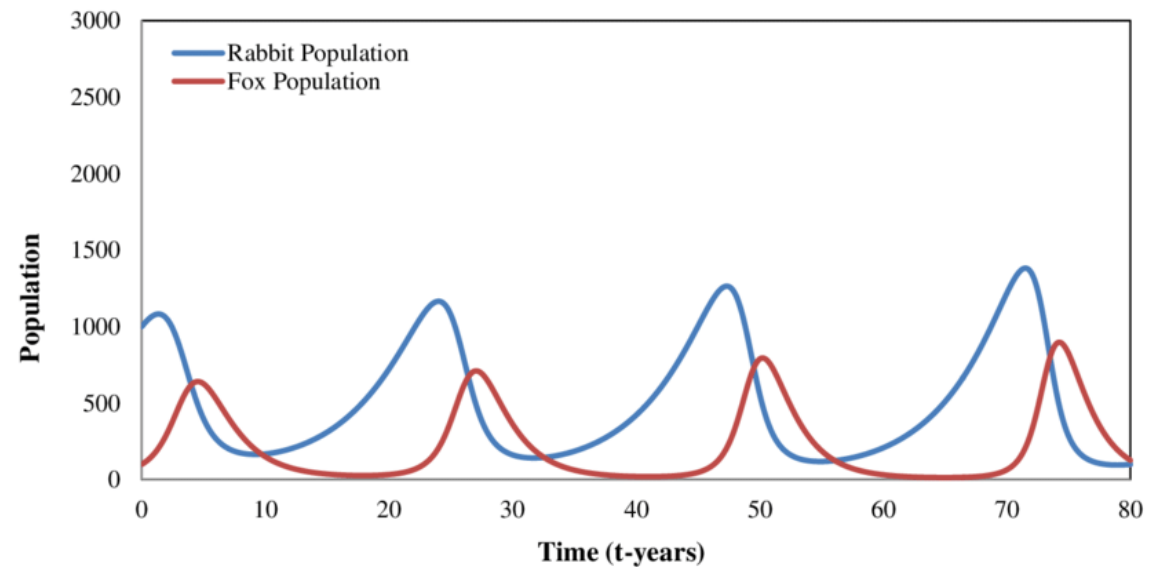
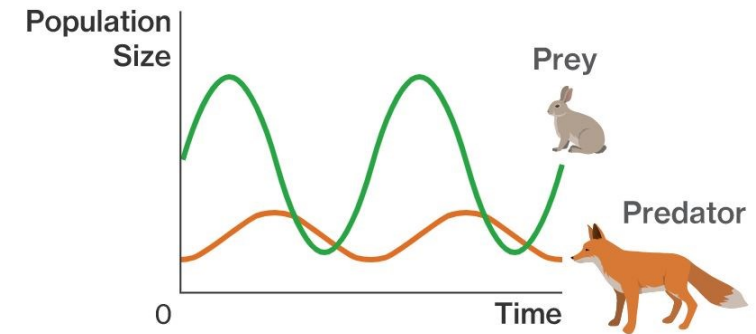
**Parameters** of the system

R: Rabbit population

F: Fox population

$r, \beta$ : birth rate

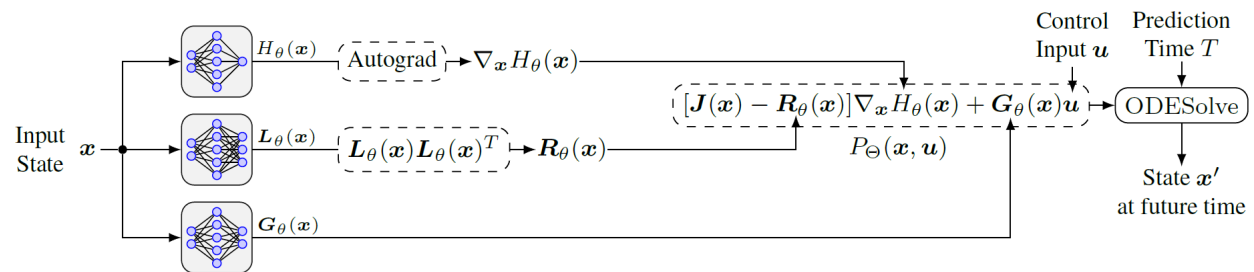
$\alpha, b$ : death rate



# Roadmap

- Hamiltonial Systems (HS) → Hamiltonian Neural Networks (HNN)
  - Closed physical Systems
- Port HS → pHNN Port Hamiltonian Neural Network
  - Real world open physical systems (with Dissipation)
- PHS Dirac → Monotonicity incremental (PHS)
  - General systems
- incremental PHS → iPHNN
- Training simple systems using PHNN

# Background



## Compositional Learning of Dynamical System Models Using PHNNs.

*Idea of composing submodels of dynamical systems is novel using DL.*

## Highlights

- Model: Ability to learn composable dynamical systems.
- Framework: Port Hamiltonian Systems Theory based.
- Implementation: Hamiltonian  $H(x)$  and dissipation  $R(x)$  NN as (MLPs).
- Training: Models are trained using ADAM.

# Hamiltonian Neural Networks

Sam Greydanus, Misko Dzamba, and Jason Yosinski (NeurIPS 2019)

## Hamiltonian neural networks

[S Greydanus, M Dzamba...](#) - Advances in neural ..., 2019 - proceedings.neurips.cc

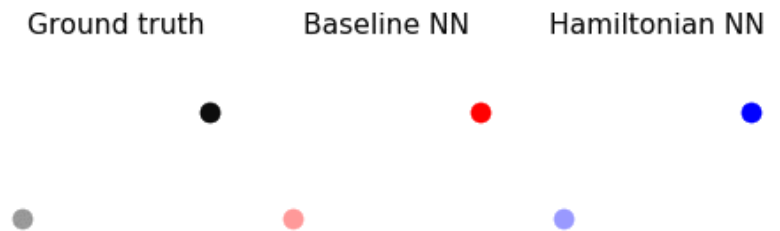
... In this paper, we draw inspiration from **Hamiltonian** mechanics, a branch of physics ... of crafting the **Hamiltonian** by hand, we propose parameterizing it with a **neural network** and ...

☆ Save 77 Cite Cited by 819 Related articles All 8 versions ⇔

Inspiration from Hamiltonian mechanics to train models that **learn and respect exact conservation laws** in an unsupervised manner.

Evaluate the model on problems where conservation of energy is important. Their model trains faster and generalizes better than a regular neural network.

Core idea: *Instead of crafting Hamiltonians by hand, they have **parameterize** them with neural networks and learn directly from data.*



Two bodies interact via a gravitational force. **The dynamics of the baseline model do not conserve total energy and quickly diverge from ground truth.** Meanwhile, the HNN learns to conserve a quantity that is close to total energy. This makes its predicted trajectories more stable.

# Why HNN?

- First problem is its **notion of discrete “time steps”** that connect neighboring states. Since time is actually continuous, a better approach would be to express dynamics as a set of differential equations and then integrate them from an initial state at  $t_0$  to a final state at  $t_1$ .
- Second problem with existing methods is that they tend not to learn exact conservation laws or invariant quantities. This often causes them to **drift away from the true dynamics** of the system as small errors accumulate.

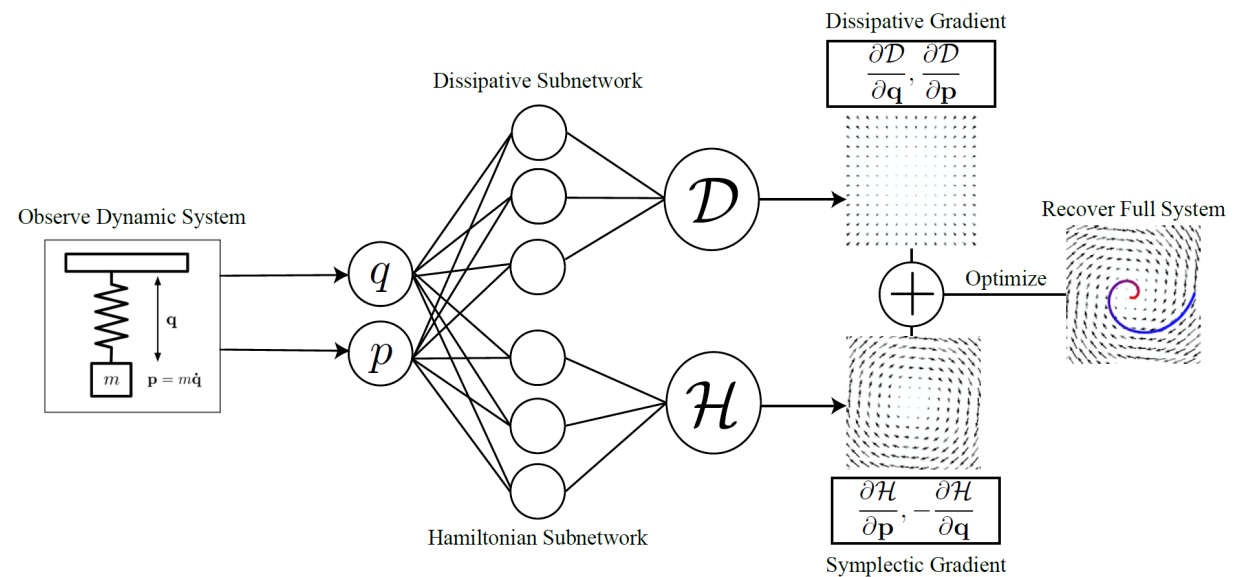
Fits well for physical systems. But what about non physical dynamical systems?

# Limitations of HNN

- Assume total energy of the system is **conserved**.
- Assume absence of friction and sources or sinks of energy.
- Real-world dynamic systems often contain **dissipative dynamics** due to friction, external forces, or measurement noise in sensor.

# Dissipative Hamiltonian Neural Networks (D-HNN)

- A model that extends HNNs to learn the decomposition of any dynamic system into its dissipative and conserved quantities.
- Trained model on a damped mass-spring systems.



Ref: [Dissipative Hamiltonian Neural Networks](#) (2022)

# Why Port-HNN Framework?

## Port-Hamiltonian Neural Networks for Learning Explicit Time-Dependent Dynamical Systems

- **Shaan A. Desai**, Marios Mattheakis, David Sondak, Pavlos Protopapas, and Stephen J. Roberts, Phys. Rev., 2021

**Hamiltonian** or **Lagrangian** neural networks predicts trajectories of physical systems. These networks generally tackle **autonomous systems** that depend implicitly on time or systems for which a control signal is known apriori.

Real world dynamical systems are **non-autonomous**, driven by time-dependent forces and experience energy dissipation. Hence, this challenge of learning from such non-autonomous systems is addressed by embedding the **port-Hamiltonian framework** that can capture energy dissipation and time-dependent control forces.

They show that the proposed **port-Hamiltonian neural network** can efficiently learn the dynamics of nonlinear physical systems of practical interest and accurately recover the underlying stationary Hamiltonian, time-dependent force, and dissipative coefficient.

[Ref: PHNNs for Learning Explicit Time-Dependent Dynamical Systems](#) (2021)

# Idea 1: incremental PHS to iPHNN

- **Basic Idea:** By replacing the composition of a Dirac structure and an energy dissipating relation by a maximal monotone relation.
- Composition of two incrementally port-Hamiltonian systems is again an iPHS based on the fact that the composition of two maximal monotone relations is again maximal monotone.
- Develop a NN based on iPHST framework using monotone relation.

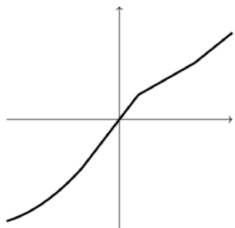
Ref: [Incrementally port-Hamiltonian systems](#) (IEEE 2013)

# iPHS

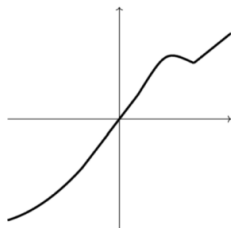
A relation  $M \subseteq F \times E$  is said to be monotone if  
 $\langle v_1 - v_2 | u_1 - u_2 \rangle \geq 0$

for all  $(u_i, v_i) \in M$  with  $i \in \{1, 2\}$ .

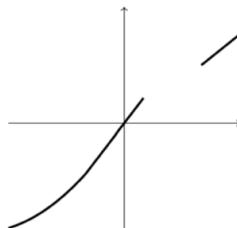
It is called maximal monotone if it is monotone and the implication  $M' \text{ is monotone and } M \subseteq M' \Rightarrow M=M'$  holds.



monotone



non-monotone



monotone but not maximal

**Definition III.1** Consider a maximal monotone relation

$$\mathcal{M} \subseteq \mathcal{F}_x \times \mathcal{E}_x \times \mathcal{F}_P \times \mathcal{E}_P$$

and a Hamiltonian  $H : \mathcal{X} \rightarrow \mathbb{R}$ . Then the dynamics of the corresponding incrementally port-Hamiltonian system is given as

$$\left( -\dot{x}(t), \frac{\partial H}{\partial x}(x(t)), f_P(t), e_P(t) \right) \in \mathcal{M} \quad (10)$$

for almost all time instants  $t$ .

It follows that the dynamics of incrementally port-Hamiltonian systems are characterized by the following ‘dissipation-like inequality’:

$$\begin{aligned} & \left\langle \frac{\partial H}{\partial x}(x_1(t)) - \frac{\partial H}{\partial x}(x_2(t)) \mid \dot{x}_1(t) - \dot{x}_2(t) \right\rangle \\ & \leq \langle e_P^1(t) - e_P^2(t) \mid f_P^1(t) - f_P^2(t) \rangle \end{aligned} \quad (11)$$

is satisfied for almost all times  $t$  for any trajectories

$$\left( -\dot{x}_i(t), \frac{\partial H}{\partial x}(x_i(t)), f_P^i(t), e_P^i(t) \right) \in \mathcal{M}$$

with  $i \in \{1, 2\}$ .

## Idea 2: Port HS and Monotonicity

- Theory and proof connecting PHS and Monotonicity.
- iPHS extended to maximal cyclically monotone relations.
- A full-fledged theory of composition of maximally monotone relations.
- Connects the theory of incrementally port HS to convex optimization.
- Equilibrium of interconnections of incrementally PHS defined by maximal cyclically monotone relations can be computed by convex optimization.

Ref: [Port-Hamiltonian systems and monotonicity \(2022\)](#)

# Idea 3: Port HS Approach to NN Training

## Conclusions

- A new perspective on the process of neural network optimization.
- Design objective function and parameter training dynamics in such a way that the neural network itself behaves as an autonomous Port-Hamiltonian system.

## Challenge

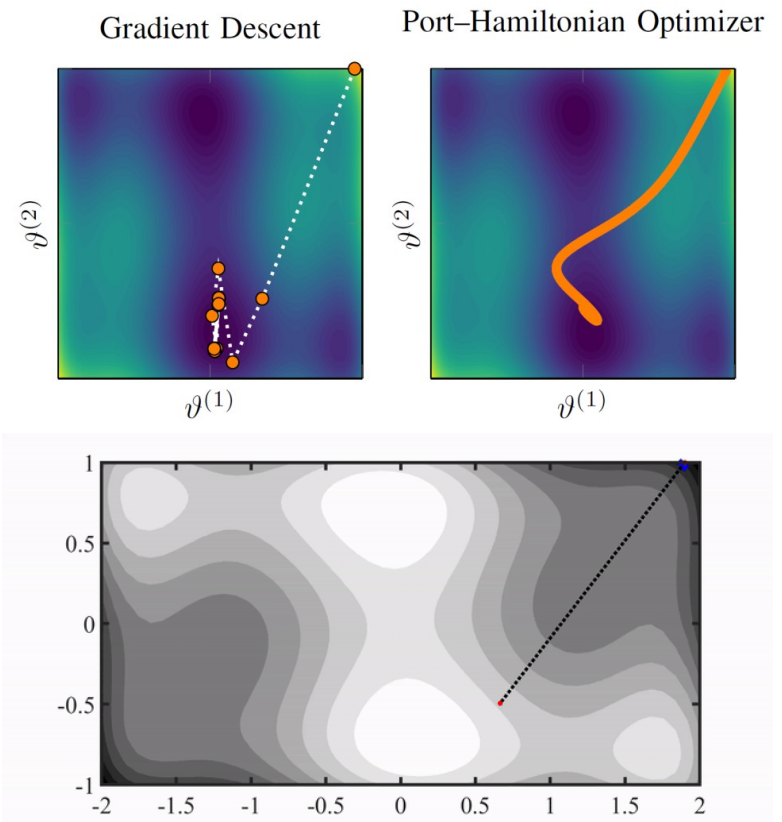
- Escaping from the saddle points due to the non-convexity and high-dimensionality of the optimization problem.

**Idea: Can we consider PHS for training a NN?**

Ref: [Port-Hamiltonian Approach to Neural Network Training](#) (2019)

15-03-2024

CS6360: Assignment - 4: Extension Ideas for PHNN



14

# PHNN: Proof of Concept

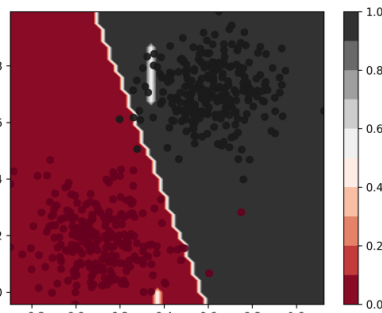
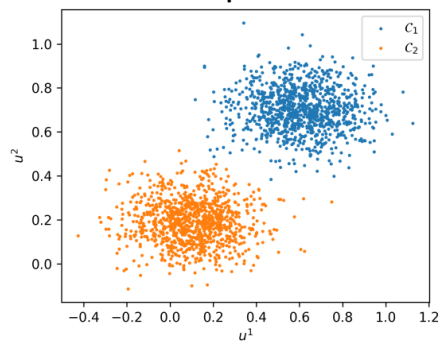
Neural network model: 
$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} w_{11}u_1 + w_{12}u_2 + w_{13} \\ w_{21}u_1 + w_{22}u_2 + w_{23} \end{bmatrix}$$

Weights Dynamics

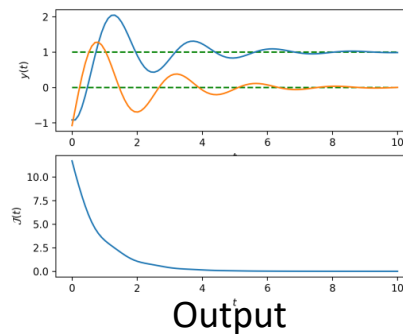
$$\begin{bmatrix} \dot{w} \\ \ddot{w} \end{bmatrix} = \begin{bmatrix} 0 & I_n \\ -I_n & -B \end{bmatrix} \begin{bmatrix} \partial_w \mathcal{J} \\ \partial_{\dot{w}} \mathcal{J} \end{bmatrix}$$

$$\Leftrightarrow \dot{x} = \begin{bmatrix} 0 & I_n \\ -I_n & -B \end{bmatrix} (\partial_x \mathcal{J})^\top$$

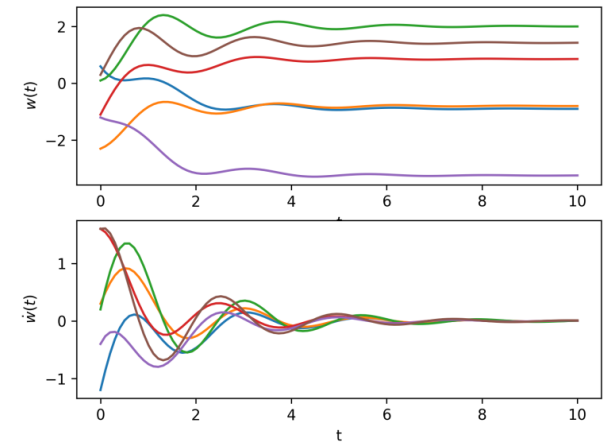
Sample Data



Decision Boundary Plot



ODE Dynamics



Model Dynamics

